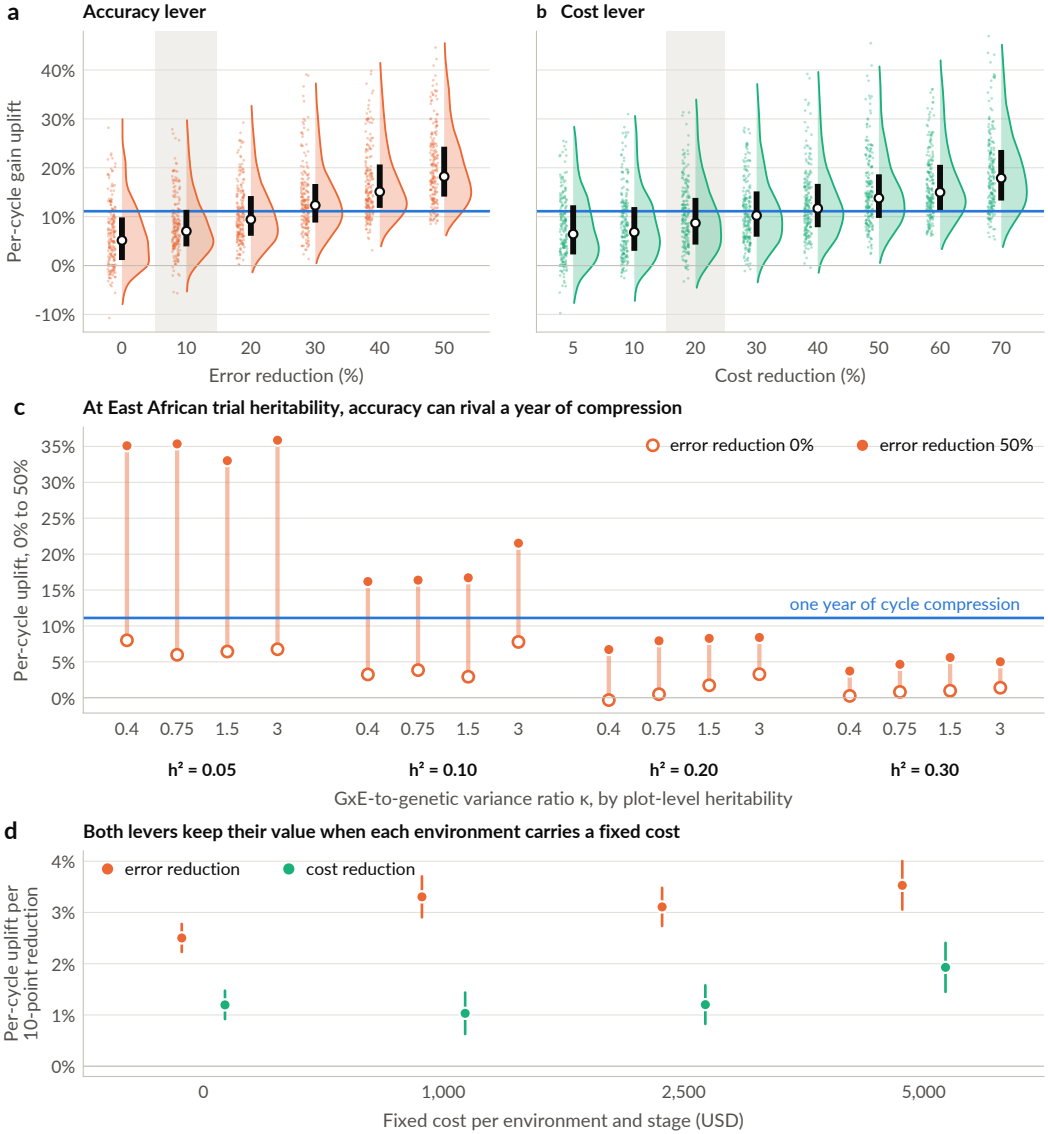

a
Accuracy lever
b Cost lever
Per-cycle gain uplift
40%
30%
20%
10%
0%
-10%
0
10
20
30
40
50
Error reduction (%)
5
10
20
30
40
50
60
70
Cost reduction (%)
c
At East African trial heritability, accuracy can rival a year of compression
Per-cycle uplift, 0% to 50%
35%
30%
25%
20%
15%
10%
5%
0%
error reduction 0%
error reduction 50%
one year of cycle compression
0.4
0.75
1.5
3
h² = 0.05
h² = 0.10
h² = 0.20
h² = 0.30
GxE-to-genetic variance ratio κ, by plot-level heritability
d
Both levers keep their value when each environment carries a fixed cost
Per-cycle uplift per 10-point reduction
4%
3%
2%
1%
0%
error reduction
cost reduction
0
1,000
2,500
5,000
Fixed cost per environment and stage (USD)